

MA1513

Chapter 1: Linear System and Matrix Algebra

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1.1 Systems of Linear Equations

In this section, we introduce the most fundamental object in Linear Algebra – System of Linear Equations (or Linear System), which play an important role in the mathematical modelling of many engineering problems and other real life applications.

1.1.1 Linear Equations

In general, a linear equation in n variables is given by this algebraic expression:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here x_1 to x_n denote the variables, while a_1 to a_n , and b represent some fixed numbers.

Specifically, a_1 to a_n are called the coefficients corresponding to the variables, and b is called the constant term.

Examples:

(a) $x + 3y = 7$

(b) $y = x - 0.5z + 4.5$

(c) $x_1 + x_2 + \cdots + x_n = 1$

(d) $x_1 + 2x_2 + 2x_3 + x_4 = x_5$

(a) is a linear equation in two variables x and y . This represents a line in the xy plane.

(b) is a linear equation in three variables x , y , z . This represents a plane in the xyz space. (Note that the x and z terms are on the right side of the equation. In any equation, we are free to move the various terms between the left and right side. But the standard form is to place all the variables on the left, and the constant term on the right as such: $-x + y + 0.5z = 4.5$.)

In (c), instead of representing the variables using regular letters like x , y , z , we use the indexed notation x_1 , x_2 , ..., x_n . This is common when there are many variables in the equation.

(d) also uses indexed notation. There are 5 variables x_1 to x_5 . Again, we can bring the variable x_5 to the left side to get the standard form.

The key feature of a linear equation is that different variables in the equation can only be combined using addition or subtraction.

1.1.2 Non-linear Equations

Examples:

(a) $xy = 2$

(b) $\sin(\theta) + \cos(\phi) = 0.2$

(c) $x_1^2 + x_2^2 + \dots + x_n^2 = 1$

(d) $x = e^y$

(a) is not linear because it involves the multiplication of the two variables x and y .

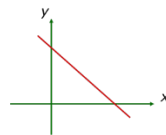
(b) is not linear in the variables θ and ϕ , as the sine and cosine functions are being applied to the variables.

(c) involves the square of each variable x_1 to x_n , and hence is not linear.

(d) is not linear because the exponential function is applied to the variable y .

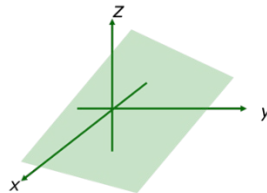
1.1.3 Geometrical Representation

If a linear equation has two variables x and y such as $ax + by = c$, then it represents a line in the xy -plane.



(Note that in this course, when we use the word “line”, we always refer to straight line, rather than a line that curves or bends.)

If a linear equation has three variables x , y and z such as $ax + by + cz = d$, then it does not represent a line in the xyz -space. Rather, it represents a flat plane in the xyz -space.



A linear equation always represent objects that are straight or flat. Furthermore, such an equation always represent an object that is “one dimension smaller” than that of the space it is sitting in.

If a linear equation $ax + by + cz + dw = e$ has four variables (x , y , z and w), it does not have any geometrical interpretation. This is because with 4 variables, it means we need a 4-dimensional space to hold the three dimensional object. This is something beyond visualization.

Nevertheless, linear equations with more than 3 variables still have many real life applications in math modelling.

1.1.4 System of Linear Equations

When a few linear equations are put together, they form a system of linear equations.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

The equations must have the same set of variables in common. Such a system is also called a linear system.

From the indexed notation, we see that there are m linear equations in the system, and n common variables x_1 to x_n .

Again, we denote the coefficients and constant terms by the a's and the b's respectively.

To better track the coefficients in the system, we have used a "double indexed" notation. For example, a_{12} refers to the coefficient of x_2 in the 1st equation, and so on.

1.1.5 Solutions of Linear System

Given a linear system, one of the problems we would like to solve is to find solutions to the system. Specifically, for a system with n variables say x_1 to x_n , we want to find n (real) numbers, say $s_1, s_2, \dots, s_n$ that satisfy all the equations simultaneously. When this happens, we say s_1 to s_n is a solution of the system and write it in the following way to specify which number correspond to which variable:

$$x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$$

This is known as a particular solution of the linear system.

Example: Linear System with more than 1 solution

$$\begin{cases} 4x_1 - x_2 + 3x_3 = -1 \\ 3x_1 + x_2 + 9x_3 = -4 \end{cases}$$

$x_1 = 1, x_2 = 2, x_3 = -1$ is a (particular) solution, since it satisfies both equations in the system.

$x_1 = 1, x_2 = 8, x_3 = 1$ is not a solution, since it satisfies the first equation but not the second one.

A linear system need not have just one solution. You can check that $x_1 = 0, x_2 = -1/4, x_3 = -5/12$ is another particular solution of the system.

In fact, we shall see later that when a linear system has more than 1 solution, it will have infinitely many solutions.

1.1.6 Example: Linear System with no solution

With the constraint of each linear equation in a system, it is highly possible for a random system to have no solution at all.

$$\begin{cases} x + y = 4 \\ 2x + 2y = 6 \end{cases}$$

This system has no solution. Observe the second equation can be simplified as $x + y = 3$. Then we see that the first equation requires the sum of the two variables to be 4, while the second equation requires the sum to be 3. In other words, the two equations are contradicting each other, or not consistent with each other, hence it is impossible to find a solution.

In general, when we have three or more equations in a linear system, it may not be easy to tell whether the equations are inconsistent by observation. We will see later how to determine the consistency of a general linear system.

1.1.7 Number of Solutions of Linear Systems

An important fact about the number of solutions that any linear system can have:

There are only three possibilities:

- no solution
- exactly one solution
- infinitely many solutions

For the first case, we call a system with no solution as an inconsistent system, as there are inconsistencies among the equations in the system.

For the second and third cases, the system has at least one solution, and we say the system is consistent.

In our earlier example

$$\begin{cases} 4x_1 - x_2 + 3x_3 = -1 \\ 3x_1 + x_2 + 9x_3 = -4 \end{cases}$$

we have found two solutions. Since the system cannot have just two solutions, we conclude that it must belong to the third case, i.e. it has infinitely many solutions, even though we have not explicitly found them. We shall see later how to find all the solutions for the infinite case.

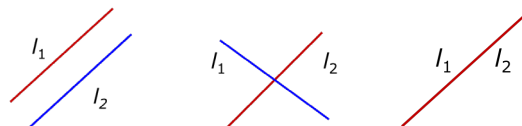
1.1.8 2 by 2 Linear Systems

The reason that there are only three possible cases for the number of solutions for any system is because it is linear. The 2 x 2 system with two equations in two variables will provide some insight to this peculiar property.

$$\begin{cases} a_1x + b_1y = c_1 & (l_1) \\ a_2x + b_2y = c_2 & (l_2) \end{cases}$$

Geometrically the two equations represent two lines l_1 and l_2 in the xy-plane.

There are three possible relative positions for the two lines:



In the first diagram, the two lines l_1 and l_2 are parallel, so they have no intersection. This means the system has no solution.

In the second diagram, l_1 and l_2 are not parallel, so they will intersect at exactly one point. This corresponds to exactly one solution for the system.

In the third diagram, the two lines overlap, so l_1 and l_2 represent the same line. This means every point on the line is part of the intersection. Hence the system has infinitely many solutions.

1.1.9 Homogeneous Linear Systems

We say a system is homogeneous if the system looks like this:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0 \end{cases}$$

Notice that it looks almost like an ordinary linear system, except that the constant terms are all 0.

On the other hand, if some of the constant terms are not zero, we say the system is non-homogeneous.

1.1.10 Trivial Solution of Homogeneous System

There is an obvious solution to the homogeneous system. When we substitute all the variables with 0, they clearly satisfy all the equations, and hence

$$x_1 = 0, x_2 = 0, \dots, x_n = 0$$

is a solution of the system. We call this the trivial solution of the homogeneous system. In particular, a homogeneous system is always consistent regardless of the coefficients.

It is possible for a homogeneous system to have only the trivial solution, or it might have other additional solutions. These additional solutions are called non-trivial solutions. Here's an example of a homogeneous system that has more than one solution.

$$\begin{cases} x_1 - x_2 + x_3 = 0 \\ x_1 + x_2 + 3x_3 = 0 \end{cases}$$

First of all, it has the trivial solution $x_1 = 0, x_2 = 0, x_3 = 0$. Check that $x_1 = 2, x_2 = 1, x_3 = -1$ is another solution. So this is a non-trivial solution to the system, and we know this implies the system in fact has infinitely many solutions.

1.1.11 Number of Solutions of Homogeneous Systems

Here are two properties about the number of solutions of homogeneous systems.

1. A homogeneous system of linear equations has either only the trivial solution or infinitely many solutions in addition to the trivial solution.
2. A homogeneous system of linear equations with more variables than equations has infinitely many solutions.

Examples:

$$\begin{cases} a_1x + b_1y + c_1z = 0 \\ a_2x + b_2y + c_2z = 0 \end{cases}$$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 = 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 = 0 \end{cases}$$

The homogeneous system on the left has 2 equations and 3 variables; and the system on the right has 3 equations and 4 variables. By the second property above, regardless of what their actual coefficients are, we can conclude that both systems have infinitely many solutions.

1.1.12 Trivial Vs Non-trivial Solutions

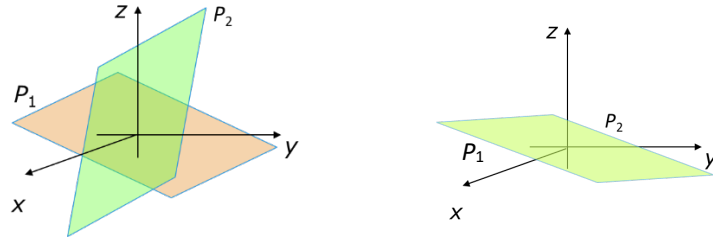
Consider the 2×3 homogeneous system, with 2 equations and 3 variables.

$$\begin{cases} a_1x + b_1y + c_1z = 0 & (P_1) \\ a_2x + b_2y + c_2z = 0 & (P_2) \end{cases}$$

Recall these two equations represent two planes in the xyz -space. The constant term being zero, means the planes contain the origin. So the two planes intersect at least the origin, which corresponds to the trivial solution.

But the property in the previous segment tells us that the system has more than just the trivial solution.

There are two possible relative positions of the two planes as shown in the diagrams below.



First possibility (left diagram) is the two planes intersect. We see that they must intersect at a line, which consists of infinitely many points. They are the origin, which represent the trivial solution, and all other points, representing the non-trivial solutions. Hence this homogeneous system has infinitely many solutions.

The second possibility (right diagram) is when the two planes overlap each other completely. This time we have the origin representing the trivial solution, and infinitely many other points on the plane representing the non-trivial solutions. Hence there are again infinitely many solutions for the homogeneous system.

1.2 Solving System of Linear Equations

When we are given a linear system, how do we determine whether it has a solution? If it does, how do we find all the solutions in a systematically way? This section provides the first step towards answering these questions.

1.2.1 Augmented Matrix

First, we use a slightly simpler notation to represent a linear system.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

This is a system with m equations and n variables.

We take all the coefficients and constant terms, and arrange them in a rectangular array as shown.

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right)$$

This array has m horizontal rows, corresponding to the m equations of the system, and $n+1$ vertical columns, corresponding to the n variables and the constant terms. Note that, to keep the notation simple, the variables, the plus and equal signs are not displayed in the array. Instead, a vertical line is drawn to separate the coefficients and the constant terms. We call this the augmented matrix of the linear system.

1.2.2 Elementary Row Operations

Next we reduce the augmented matrix to a simpler form, so that it becomes easier to solve the linear system. The way to do this is by performing some operations on the rows of the matrix. There are three types of row operations, namely

- Multiplying a row by a nonzero constant;
- Interchanging two rows;
- Adding a multiple of a row to another row.

These are the three basic operations on the rows of a matrix, which we called the elementary row operations.

Example: Elementary Row Operations

For the first type of row operation, multiply the first row of the left augmented matrix by the number 3, we will get the augmented matrix on the right.

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 3 & 3 & 6 & 27 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right)$$

Notice that all the entries in the first row get multiplied by 3.

For the second type of row operation, we interchange the second and third row of the left augmented matrix to get the right.

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 3 & 6 & -5 & 0 \\ 2 & 4 & -3 & 1 \end{array}\right)$$

For the third type of row operation, we want to add 2 times of the first row to the second row. First, we multiply the first row of the left augmented matrix by 2 to get $(2 \ 2 \ 4 \mid 18)$. Then we add this to the second row of the same matrix, by adding the corresponding entries. The resulting matrix looks like this.

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 4 & 6 & 1 & 19 \\ 3 & 6 & -5 & 0 \end{array}\right)$$

Note that in the resulting matrix, only the second row is changed; the first row will remain the same as the original matrix.

1.2.3 Notation of Elementary Row Operations

Let us denote the rows of an augmented matrix to be operated on using the indexed notation R_i . So R_1 represents the first row, R_3 represents the third row and so on. With that, we can now represent the three row operations as follow:

First type: cR_i (multiplying the i^{th} row by the constant c)

Second type: $R_i \leftrightarrow R_j$ (interchanging the positions of the i^{th} and j^{th} rows)

Third type: $R_i + cR_j$ (adding c times the j^{th} row to the i^{th} row)

In the last notation, it means the i^{th} row will be changed, but the j^{th} row remains unchanged after the operation.

1.2.4 Example: Solving Linear System

Take note of the action taken on the left hand side (the linear system), and the corresponding row operations being performed on the right side (the augmented matrix).

In the first step below, we add -2 times of equation 1 to equation 2, and change equation 2 to equation 4 as shown. Correspondingly, the second row of the augmented matrix is changed accordingly by adding -2 times of first row of the augmented matrix to the second row as shown in the diagram.

$$\begin{array}{l} \left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) \\ 2x - 2y + 2z = 4 & (2) \\ 3x + 9y = 3 & (3) \end{array} \right. & \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & -2 & 2 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right) \\ \left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) \\ -x - 4y - 4z = 4 & (4) \\ 3x + 9y = 3 & (3) \end{array} \right. & \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right) \end{array}$$

$R_2 + (-2)R_1$

In this next step, we change equation 3 to equation 5. This corresponds to the row operation of adding -3 times of the first row of the matrix to the third row.

$$\left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) & \\ -4y - 4z = 4 & (4) & \\ 3x + 9y = 3 & (3) & \end{array} \right. \quad \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right)$$

$$\xrightarrow{R_3 + (-3)R_1} \left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) & \\ -4y - 4z = 4 & (4) & \\ 6y - 9z = 3 & (5) & \end{array} \right. \quad \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 6 & -9 & 3 \end{array} \right)$$

Next, we change equation 5 to equation 6. This corresponds to the row operation of adding 6/4 times of the second row of the matrix to the third row.

$$\left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) & \\ -4y - 4z = 4 & (4) & \\ 6y - 9z = 3 & (5) & \end{array} \right. \quad \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 6 & -9 & 3 \end{array} \right)$$

$$\xrightarrow{R_3 + (6/4)R_2} \left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) & \\ -4y - 4z = 4 & (4) & \\ -15z = 9 & (6) & \end{array} \right. \quad \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 0 & -15 & 9 \end{array} \right)$$

Let's summarize how the original augmented matrix has been transformed under the operations:

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & -2 & 2 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right) \xrightarrow{R_2 + (-2)R_1} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 3 & 9 & 0 & 3 \end{array} \right) \xrightarrow{R_3 + (-3)R_1} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 6 & -9 & 3 \end{array} \right) \xrightarrow{R_3 + (6/4)R_2} \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 0 & -15 & 9 \end{array} \right)$$

Observe that the augmented matrix now has a "staircase" of 0 at the bottom left corner of the array. This is the simplified form of the augmented matrix that we are looking for. It is known as the row echelon form (which will be defined more precisely later).

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 0 & -15 & 9 \end{array} \right)$$

In the corresponding linear system, we observe the number of variables in each equation decreases as we go down the row. This will facilitate us to solve the equations from bottom up, which we call back substitution.

$$\left\{ \begin{array}{lcl} x + y + 3z = 0 & (1) & \\ -4y - 4z = 4 & (4) & \\ -15z = 9 & (6) & \end{array} \right.$$

First, from (6), we can easily solve $z = -3/5$.

Substituting this value of z into (4), we get $-4y - 4(-3/5) = 4$ which gives $y = -2/5$.

Finally, substituting the values of y and z into (1), we get $x - 2/5 - 9/5 = 0$, which gives $x = 11/5$.

These three numbers give us a solution of the linear system. In fact, from the process, we see that this is the only solution of the original linear system.

1.2.5 Row Echelon Form (R.E.F.)

- In a row echelon form, any zero rows must be below all the nonzero rows.

A zero row is one whereby all the entries are 0: $(0 \ 0 \ \dots \ 0 \mid 0)$; and a nonzero row is one with some nonzero entries. For example, $(0 \ 0 \ 3 \ 0 \ 1 \mid 0)$ and $(0 \ 0 \ 0 \ 0 \ 0 \mid 2)$ are examples of non-zero rows.

If we go from left to right on each nonzero row, the first nonzero entry is called the leading entry. So the leading entry of $(0 \ 0 \ 3 \ 0 \ 1 \mid 0)$ is 3, and the leading entry of $(0 \ 0 \ 0 \ 0 \ 0 \mid 2)$ is 2.

In the diagram below, the symbol $\otimes$ represents the leading entry of each non-zero row. (Note that entries * on the right of the leading entry can be either zero or non-zero.)

$$\left(\begin{array}{cccccc|c} 0 & \otimes & * & * & * & * & * \\ 0 & 0 & 0 & \otimes & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \otimes & * \\ 0 & 0 & 0 & 0 & 0 & 0 & \otimes \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The most important feature of a row echelon form is that

- the number of zeros on the left of the leading entries is strictly increasing as we go down the row, creating a staircase of zeros at the bottom left of the matrix.
(Note that the width of each stair need not be the same.)

We are interested in the columns of a row echelon form that contain the leading entries. We call these columns the pivot columns.

$$\left(\begin{array}{cccccc|c} 0 & \otimes & * & * & * & * & * \\ 0 & 0 & 0 & \otimes & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \otimes & * \\ 0 & 0 & 0 & 0 & 0 & 0 & \otimes \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

In the above augmented matrix, the pivot columns are the 2nd, 4th, 6th and 7th columns. The remaining columns in the matrix are called the non-pivot columns.

The pivot and non-pivot columns in the row echelon form play an important role in determining the number of solutions of a linear system.

1.2.6 Example: System with Infinitely Many Solutions

$$\left\{ \begin{array}{l} 2x_1 + 2x_2 + x_3 - 2x_4 = 2 \quad (1) \\ + x_2 + x_3 + x_4 = 3 \quad (2) \\ + 2x_4 = 4 \quad (3) \end{array} \right. \quad \left(\begin{array}{cccc|c} 2 & 2 & 1 & -2 & 2 \\ 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 2 & 4 \end{array} \right)$$

This is a system that is already reduced. In other words, its augmented matrix is in row echelon form.

This system has 4 variables and 3 equations. In other words, there are more variables than equations. We need to set some variables as free parameters. The variables to be set as parameters are those that correspond to non-pivot columns in the row echelon form.

In our example, the third column is a non-pivot column, and it corresponds to variable x_3 . So we shall set x_3 as a free parameter, say t , and write

$$x_3 = t \text{ where } t \in \mathbb{R}.$$

In other words, x_3 can represent any real number.

We shall substitute all the variable x_3 in the system with the parameter t and perform back-substitution as before:

From (3), we solve $x_4 = 2$.

Substituting x_4 by 2 in the system, we can then solve (2) for x_2 in terms of t : $x_2 = 1 - t$.

Substituting x_2 by $1 - t$ in the system, we can finally solve $x_1 = 2 + (1/2)t$.

By putting all these together, we can write the solution in the following way:

$$\begin{cases} x_1 = 2 + \frac{1}{2}t \\ x_2 = 1 - t \\ x_3 = t \\ x_4 = 2 \end{cases}$$

The parameter t represents any real number, and therefore gives infinitely many solutions of the linear system. In fact, this will give all the possible solutions of the system, and hence it is called a general solution of the system.

1.2.7 Solutions of Linear Systems

We can determine how many solutions a system has from the row echelon form of the augmented matrix of the system.

- (i) When every column of the row echelon form is a pivot column, except the last column, the system will have exactly one solution.

Here is an example we have seen earlier. (The columns in blue are pivot columns.)

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -4 & -4 & 4 \\ 0 & 0 & -15 & 9 \end{array} \right)$$

- (ii) When there is at least one non-pivot column in the row echelon form other than the last column, the system will have infinitely many solutions.

This is an example we have seen in the previous segment. (The column highlighted is the non-pivot column.)

$$\left(\begin{array}{ccc|c} 2 & 2 & 1 & -2 & 2 \\ 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 2 & 4 \end{array} \right)$$

Recall that the non-pivot columns (can be more than one) correspond to variables being set as free parameters, and hence giving infinitely many solutions.

What about the case of no solution? Can we tell from the row echelon form? The answer is yes.

- (iii) When the very last column of the row echelon form is a pivot column, the system will have no solution.

1.2.8 Example: System with No Solution

Suppose a linear system has the augmented matrix on the left below and we perform some elementary row operations to reduce it to a row echelon form.

$$\left(\begin{array}{ccc|c} 3 & 1 & 4 \\ 0 & 2 & 1 \\ 3 & 3 & 6 \end{array}\right) \xrightarrow{R_3 + (-1)R_1} \left(\begin{array}{ccc|c} 3 & 1 & 4 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{array}\right) \xrightarrow{R_3 + (-1)R_2} \left(\begin{array}{ccc|c} 3 & 1 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{array}\right)$$

Convert this back to equation form,

$$\begin{cases} 3x_1 + x_2 = 4 \\ 2x_2 = 1 \\ 0 = 1 \end{cases}$$

we see that the third equation is $0 = 1$, which is impossible. This suggests that the system is inconsistent without solution. The corresponding row in the row echelon form (rightmost augmented matrix above) is $(0 \ 0 \ | \ 1)$, which means that 1 is a leading entry, and hence the last column in the row echelon form is a pivot column as indicated in the diagram.

1.2.9 Non-homogeneous VS Homogeneous System

This is an earlier example of linear system, which we denote by (A).

$$(A) \quad \begin{cases} 2x_1 + 2x_2 + x_3 - 2x_4 = 2 \\ x_2 + x_3 + x_4 = 3 \\ 2x_4 = 4 \end{cases}$$

We have obtained the general solution of (A). By setting t with any fix value, we get a particular solution of the system. For example, if we let $t = 0$, then we get a particular solution as shown below.

$$\begin{array}{lcl} \text{general solution} & & \text{a particular solution} \\ \left\{ \begin{array}{l} x_1 = 2 + \frac{1}{2}t \\ x_2 = 1 - t \\ x_3 = t \\ x_4 = 2 \end{array} \right. & \xrightarrow{\text{Set } t=0} & \left\{ \begin{array}{l} x_1 = 2 \\ x_2 = 1 \\ x_3 = 0 \\ x_4 = 2 \end{array} \right. \end{array}$$

Write down the associated homogeneous system of (A) as shown below, and denote it by (B):

$$(B) \quad \begin{cases} 2x_1 + 2x_2 + x_3 - 2x_4 = 0 \\ x_2 + x_3 + x_4 = 0 \\ 2x_4 = 0 \end{cases}$$

Note that (A) and (B) are almost the same except that the constant terms in (B) are all 0.

By going through the same process (back substitution) as before, we can find the general solution of the homogeneous system:

$$\text{general solution of (B)} \quad \left\{ \begin{array}{l} x_1 = \frac{1}{2}t \\ x_2 = -t \\ x_3 = t \\ x_4 = 0 \end{array} \right.$$

Compare the two general solutions, we notice the similarity of the parameter terms between the two systems. Furthermore, we also observe the relations between the constant terms of the general solution of (A) and the particular solution of (A) above.

From this example, we can formulate the following relationship:

$$(**) \quad \text{general solution of (A)} = \text{general solution of (B)} + \text{a particular solution of (A)}$$

The general solution of the non-homogeneous system (A) is the sum of the general solution of the associated homogeneous system (B) and a particular solution of (A).

In fact, given any non-homogeneous system and its associated homogeneous system, if the non-homogeneous system is consistent, the same relationship (**) between their general solution holds.

Note that the particular solution in (**) can be obtained by substituting any value for the parameters, i.e. it is not necessary to set $t = 0$.

In other words,

$$(\Delta\Delta) \text{ general solution of (A) }^* = \text{general solution of (B)} + \text{another particular solution of (A)}$$

* The general solution of (A) in ($\Delta\Delta$) may not look the same as the general solution of A in (**). But both of them will give all possible solutions of (A).

1.3 Gaussian Elimination

This section illustrates a systematic approach to reduce the augmented matrix of the system to a row echelon form. The method Gaussian Elimination (G.E.) that we are going to introduce is named after the great mathematician Carl Friedrich Gauss. The process is a series of elementary row operations performed according to a set of instruction. The objective is to reduce an augmented matrix to a row echelon form.

1.3.1 Gaussian Elimination

We will use the following two augmented matrices as examples to illustrate the steps.

$$\begin{array}{cc} \text{Example A} & \text{Example B} \\ \left(\begin{array}{cccc} 0 & 3 & \dots & \dots \\ 1 & -2 & \dots & \dots \\ 4 & 0 & \dots & \dots \end{array} \right) & \left(\begin{array}{cccc} 0 & 3 & 1 & \dots \\ 0 & 2 & -3 & \dots \\ 0 & -1 & 6 & \dots \end{array} \right) \end{array}$$

Step 1

- ❖ Locate the leftmost column that does not consist entirely of zero.

$$\begin{array}{cc} \text{Example A} & \text{Example B} \\ \left(\begin{array}{cccc} 0 & 3 & \dots & \dots \\ 1 & -2 & \dots & \dots \\ 4 & 0 & \dots & \dots \end{array} \right) & \left(\begin{array}{cccc} 0 & 3 & 1 & \dots \\ 0 & 2 & -3 & \dots \\ 0 & -1 & 6 & \dots \end{array} \right) \\ \text{the first} & \text{the first} \\ \text{nonzero} & \text{nonzero} \\ \text{column} & \text{column} \end{array}$$

Step 2

- ❖ Interchange the top row with another row, if necessary, to bring a nonzero entry to the top of the column found in Step 1.

$$\begin{array}{cc} \text{Example A} & \text{Example B} \\ \left(\begin{array}{cccc} 0 & 3 & \dots & \dots \\ 1 & -2 & \dots & \dots \\ 4 & 0 & \dots & \dots \end{array} \right) & \left(\begin{array}{cccc} 0 & 3 & 1 & \dots \\ 0 & 2 & -3 & \dots \\ 0 & -1 & 6 & \dots \end{array} \right) \\ \text{Interchange} & \text{No action is needed} \\ \text{the 1st row with} & \\ \text{the 2nd row.} & \end{array}$$

For example A, the top entry is 0, so we perform the second type of row operation, namely interchanging the 1st row with another row with nonzero entry. In this case, we can interchange it with the second row $R_1 \leftrightarrow R_2$. In fact, it is also possible to interchange the 1st row with the 3rd row.

For example B, as the top entry is nonzero, it is not necessary to interchange row.

Step 3

- ❖ For each row below the top row, add a suitable multiple of the top row to it so that the entry below the leading entry of the top row becomes zero.

$$\begin{array}{cc} \text{Example A} & \text{Example B} \\ \left(\begin{array}{cccc} 1 & -2 & \dots & \dots \\ 0 & 3 & \dots & \dots \\ 4 & 0 & \dots & \dots \end{array} \right) & \left(\begin{array}{cccc} 0 & 3 & 1 & \dots \\ 0 & 2 & -3 & \dots \\ 0 & -1 & 6 & \dots \end{array} \right) \\ \text{Add } -4 \text{ times of the} & \text{Add } -2/3 \text{ times of the 1st row to the} & \text{Add } 1/3 \text{ times of the} \\ \text{1st row to the 3rd} & \text{1st row to the 2nd row so that the} & \text{1st row to the 3rd} \\ \text{row so that the entry} & \text{entry marked by } \bigcirc & \text{row so that the entry} \\ \text{marked by } \bigcirc & & \text{marked by } \bigcirc \\ \text{becomes 0.} & & \text{becomes 0.} \end{array}$$

In example A, there is only one nonzero entry 4 in the third row. So only one row operation is needed. We perform the third type of operation $R_3 - 4R_1$.

In example B, there are two nonzero entries 2 and -1 below the top entry. So we need two row operations to turn them into 0. We perform $R_2 - (2/3)R_1$ and $R_3 + (1/3)R_1$.

(Remember, we are trying to build a staircase of zero at the bottom left corner of the matrices.)

Step 4

- ❖ Now cover the top row in the matrix and begin again with Step 1 applied to the submatrix that remains.

Example A	Example B
$\begin{pmatrix} 1 & -2 & \dots & \dots \\ 0 & 3 & \dots & \dots \\ 0 & 8 & \dots & \dots \end{pmatrix}$	$\begin{pmatrix} 0 & 3 & 1 & \dots \\ 0 & 0 & -11/3 & \dots \\ 0 & 0 & 19/3 & \dots \end{pmatrix}$

In example A, after covering the top row, the next nonzero column is the second column. In example B, after covering the top row, the next nonzero column is the third column. We repeat step 1 to step 3 to both matrices.

Final Step

- ❖ After each cycle, we cover one more row, and repeat the process, until the matrix is reduced to a row-echelon form.

Example A	Example B
$\begin{pmatrix} 1 & -2 & \dots & \dots \\ 0 & 3 & \dots & \dots \\ 0 & 0 & \dots & \dots \end{pmatrix}$	$\begin{pmatrix} 0 & 3 & 1 & \dots \\ 0 & 0 & -11/3 & \dots \\ 0 & 0 & 0 & \dots \end{pmatrix}$

Note that the final matrices that we get in both examples are in row echelon form. This completes the Gaussian elimination.

1.3.2 Reduced Row Echelon Form (R.R.E.F.)

- A row echelon form is said to be a reduced row echelon form, if in each of its pivot columns, the leading entry is 1 and all other entries are 0.

Let us illustrate this with our diagram. For a row echelon form, the leading entries can be any nonzero numbers. But for a reduced row echelon form, the leading entries are all equal to 1. Furthermore, in the pivot columns, all other non-leading entries are 0.

$\left(\begin{array}{cccccc c} 0 & \otimes & * & * & * & * & * \\ 0 & 0 & 0 & \otimes & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \otimes & * \\ 0 & 0 & 0 & 0 & 0 & 0 & \otimes \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$	$\left(\begin{array}{cccccc c} 0 & 1 & * & 0 & * & 0 & 0 \\ 0 & 0 & 0 & 1 & * & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$
Row echelon form	Reduced row echelon form

In some applications, it is necessary for us to reduce a matrix to this specific reduced row echelon form. In order to do that, we need to extend the Gaussian elimination to include additional steps. This modified version of the procedure is called Gauss-Jordan elimination.

1.3.3 Gauss-Jordan Elimination (G.J. E.)

Step A: Multiply a suitable constant to each row so that all the leading entries becomes 1.

We use a different example here for illustration. This matrix is in row echelon form (the blank space on the bottom left corner represent a staircase of zero entries), but some of the leading entries are not 1.

Diagram illustrating the initial matrix in row echelon form. The matrix is shown with leading entries circled in red: 2, 1, -3, and 1. Annotations indicate the required row operations:

- Multiply this row by $1/2$ so that the leading entry becomes 1. (Pointing to the first row's leading entry 2)
- No action is needed. (Pointing to the second row's leading entry 1)
- Multiply this row by $-1/3$ so that the leading entry becomes 1. (Pointing to the third row's leading entry -3)
- No action is needed. (Pointing to the fourth row's leading entry 1)

So we need to perform type 1 elementary row operations on those rows indicated. The resulting augmented matrix is given below:

$$\begin{pmatrix} 1 & -2 & \dots & \dots & -4 & \dots \\ & 1 & \dots & \dots & 3 & \dots \\ & & \ddots & \dots & \dots & \dots \\ & & & 1 & 2 & \dots \\ & & & & 1 & \dots \end{pmatrix}$$

Step B: Start from the rightmost pivot column and work “upward” — add a suitable multiples of each row to the rows above to introduce 0 above the leading entries.

Diagram illustrating the matrix after Step B. The leading entries are circled in red: 1, 1, 3, and 1. The entry 3 in the third row is also circled in blue. Annotations indicate the required row operations:

- Add 4 times of the last row to the 1st row so the entry marked by 0 becomes 0. (Pointing to the -4 in the first row)
- Add -3 times of the last row to the 2nd row so the entry marked by 0 becomes 0. (Pointing to the 3 in the second row)
- Add -2 times of the last row to the next row so the entry marked by 0 becomes 0. (Pointing to the 2 in the third row)

A red arrow on the right points upwards, labeled "Work from bottom up".

We work from “bottom up” starting with the entry 2 right above the leading entry. For this entry, we add -2 times the bottom row to its row. Similarly, for the entry 3, we add -3 times of the bottom row to its row, which is the second row. Finally, for the entry -4, we add 4 times of the bottom row to the first row.

With that, we have completed the rightmost pivot columns. We then move on to the next pivot columns to the left and repeat the same procedure.

Diagram illustrating the matrix after moving to the next pivot column. The leading entries are circled in red: 1, 1, 1, and 1. The entry 1 in the third row is also circled in blue. Annotations indicate the required row operations:

- Apply the same process to the next pivot column on the left. (Pointing to the 1 in the third row)

A red arrow at the bottom points to the left, labeled "Work from right to left".

When this next pivot column is done, we move to the next one on the left, and so on.

Remember, for these additional steps in Gauss-Jordan elimination,

we have to work from right to left, bottom to top.

1.3.4 Example: Solving a Linear System using G.E./G.J.E.

$$\begin{cases} 2x_3 + 4x_4 + 2x_5 = 8 \\ x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9 \\ -2x_1 - 4x_2 - 5x_3 - 4x_4 + 3x_5 = 6 \end{cases}$$

Our objective is to make a comparison solve this linear system by using Gaussian elimination and Gauss Jordan elimination. We start from the augmented matrix.

$$\left(\begin{array}{ccccc|c} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right)$$

After performing Gaussian elimination, we obtain a row echelon form:

$$\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right) \xrightarrow{R_3 \mp 2R_1} \left(\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{array} \right) \xrightarrow{R_3 - 3/2R_2} \left(\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right)$$

If we continue with Gauss Jordan elimination, we will get a reduced row echelon form:

$$\xrightarrow{1/2R_2, 1/6R_3} \left(\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right) \xrightarrow{R_2 - R_3} \left(\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right) \xrightarrow{R_1 - 3R_3} \left(\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right) \xrightarrow{R_1 - 4R_2} \left(\begin{array}{ccccc|c} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right)$$

Put the row echelon form and reduced row echelon form respectively back in standard equation forms, we get the systems:

$$\begin{cases} x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9 \\ 2x_3 + 4x_4 + 2x_5 = 8 \\ 6x_5 = -24 \end{cases}$$

By Gaussian elimination

VS

$$\begin{cases} x_1 + 2x_2 - 3x_4 = -29 \\ x_3 + 2x_4 = 8 \\ x_5 = -4 \end{cases}$$

By Gauss-Jordan elimination

We observe that the system using G.J.E. is much simpler than that using G.E. Nevertheless, the number of equations and variables in both systems are the same. In both cases, we need to introduce parameters to two of the variables, since there are two more variables than equations.

If we look at the non-pivot columns (2nd and 4th columns) of the (reduced) row echelon form, the variables that we need to set as parameters are x_2 and x_4 , regardless of which method we use.

By setting $x_2 = s$ and $x_4 = t$ and substitute them into the systems, we can then solve the system in terms of these two parameters.

$$\begin{cases} x_1 + 2s + 4x_3 + 5t + 3x_5 = -9 \\ 2x_3 + 4t + 2x_5 = 8 \\ 6x_5 = -24 \end{cases}$$

$$\begin{cases} x_1 + 2s - 3t = -29 \\ x_3 + 2t = 8 \\ x_5 = -4 \end{cases}$$

For the system from G.E., we need to do back substitution to get the general solution. For the simpler system from G.J.E., we get the general solution immediately by bringing the parameters to the right hand side of the equations. We notice that the general solutions using either method are the same:

$$\begin{cases} x_1 = -29 - 2s + 3t \\ x_2 = s \\ x_3 = 8 - 2t \\ x_4 = t \\ x_5 = -4 \end{cases}$$

with parameters t and s representing any real numbers.

1.4 Matrices

We have come across the augmented matrix of a linear system in the earlier discussion. In this section, we will formally introduce the concept of matrices and some of their operations.

1.4.1 Terminologies

The singular form is called Matrix; while the plural form is called Matrices. It is represented by a rectangular array of numbers. Unlike the augmented matrix we have seen earlier, there is no separating lines to separate the columns.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

A matrix has horizontal rows and vertical columns. From the indexed notation, we see that there are m rows and n columns in this matrix $\mathbf{A}$. We call $\mathbf{A}$ an $m \times n$ matrix and refer to $m \times n$ as the size of the matrix.

The entry a_{ij} is called the ij -entry of the matrix. The double indices indicate the position of the entry, i.e. it is on the i^{th} row and j^{th} column. For example, the (1,2)-entry a_{12} is on the first row and second column, while the (2,1)-entry a_{21} is on the second row and first column.

1.4.2 Special Types of Matrices

Square matrix	number of rows and columns are the same	$\begin{pmatrix} 0 & 3 & 5 \\ 1 & -2 & 3 \\ 4 & 0 & 8 \end{pmatrix}$
Diagonal matrix	square matrix with all non-diagonal entries being zero	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$
Scalar matrix	diagonal matrix with all diagonal entries the same	$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$
Identity matrix $\mathbf{I}_n$	diagonal matrix with all diagonal entries being 1	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

This is an example of a 3×3 square matrix.

Diagonal entries of an $n \times n$ matrix refers to the (1,1)-entry, (2,2)-entry, up to (n, n)-entry. In this example, the diagonal entries are 1, 3, 2.

A scalar matrix is a special type of diagonal matrix.

We have a special notation $\mathbf{I}_n$ to denote the $n \times n$ identity matrix. So the 3×3 identity matrix here is denoted by $\mathbf{I}_3$.

Zero matrix $\mathbf{O}_{m \times n}$	all entries equal to zero	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
Symmetric matrix	k^{th} row "equal" k^{th} column for all k	$\begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & 2 \\ 0 & 2 & 2 \end{pmatrix}$
Upper triangular matrix	all entries below diagonal are zero	$\begin{pmatrix} 1 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 2 \end{pmatrix}$
Lower triangular matrix	all entries above diagonal are zero	$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 2 & 3 & 2 \end{pmatrix}$

Zero matrix need not be a square matrix. We can have zero matrices of any sizes. We have a special notation $\mathbf{O}_{m \times n}$ to denote the $m \times n$ zero matrix.

In our example, note that the first row consists of entries 1 -1 0, and the first column also consists of 1 -1 0. Similarly for the other row-column pairs.

All entries on and "above" the diagonals have no restriction, i.e. they can be zero or non-zero.

All entries on or "below" the diagonals can be zero or non-zero.

1.4.3 Matrix Operations

Let $\mathbf{A}$ and $\mathbf{B}$ be matrices of the same size, and c a real number (known as a scalar). The table below summarizes the four basic operations on matrices. The column on the right provides examples for the 2×2 case.

Matrix Addition	$\mathbf{A} + \mathbf{B}$	addition of corresponding entries of $\mathbf{A}$ and $\mathbf{B}$	$\begin{pmatrix} 0 & 3 \\ 1 & -2 \end{pmatrix} + \begin{pmatrix} 2 & 7 \\ 8 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 10 \\ 9 & 0 \end{pmatrix}$
Matrix subtraction	$\mathbf{A} - \mathbf{B}$	subtraction of corresponding entries of $\mathbf{A}$ and $\mathbf{B}$	$\begin{pmatrix} 0 & 3 \\ 1 & -2 \end{pmatrix} - \begin{pmatrix} 2 & 7 \\ 8 & 2 \end{pmatrix} = \begin{pmatrix} -2 & -4 \\ -7 & -4 \end{pmatrix}$
Scalar multiplication	$c\mathbf{A}$	multiply every entry of $\mathbf{A}$ by scalar c	$4\begin{pmatrix} 0 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 12 \\ 4 & -8 \end{pmatrix}$
Negative of matrix	$-\mathbf{A}$	attach negative sign to every entry of $\mathbf{A}$	$-\begin{pmatrix} 0 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 0 & -3 \\ -1 & 2 \end{pmatrix}$

1.4.4 Matrix Multiplication

A slightly less straight forward, but very useful operation, is matrix multiplication. We first illustrate using a simple example.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix}$$

We represent matrix multiplication of two matrices by putting them side by side. If we denote the two matrices by $\mathbf{A}$ and $\mathbf{B}$, then the notation of the product is $\mathbf{AB}$. Here we have a 2×3 matrix $\mathbf{A}$ on the left, and a 3×2 matrix $\mathbf{B}$ on the right. After the multiplication, the resulting matrix $\mathbf{AB}$ is 2×2 .

$$\mathbf{AB} = \begin{pmatrix} 1 \times 1 + 2 \times 2 + 3 \times (-1) & 1 \times 1 + 2 \times 3 + 3 \times (-2) \\ 4 \times 1 + 5 \times 2 + 6 \times (-1) & 4 \times 1 + 5 \times 3 + 6 \times (-2) \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}$$

For the (1,1)-entry, we take the first row of $\mathbf{A}$ and the first column of $\mathbf{B}$ and take the sum of the product of their respective entries as shown. For the (1,2)-entry, we take the first row of $\mathbf{A}$ and the second column of $\mathbf{B}$ and do the same. Similarly, for the (2,1)-entry, we use the second row of $\mathbf{A}$ and the first column of $\mathbf{B}$, and for the (2,2)-entry, we use the second row of $\mathbf{A}$ and the second column of $\mathbf{B}$. After simplifying, we get the resulting product on the right.

1.4.5 Multiply Matrices of Compatible Sizes

We can only multiply two matrices $\mathbf{A}$ and $\mathbf{B}$ (in the manner $\mathbf{AB}$) when the number of columns of $\mathbf{A}$ is equal to the number of rows of $\mathbf{B}$. In other words, if $\mathbf{A}$ is of size $m \times p$, then $\mathbf{B}$ must be of size $p \times n$ in order to perform the multiplication $\mathbf{AB}$. In this case, the size of the product $\mathbf{AB}$ will be $m \times n$.

To illustrate with concrete examples, we are able to multiply the pair of 2×3 and 3×2 matrices, but we cannot multiply the pair of 2×3 and 2×3 matrices.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix}$$

Matrix multiplication
can be performed

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix}$$

Matrix multiplication
cannot be performed

1.4.6 Properties of Matrix Multiplication

1. Matrix multiplication is not commutative: $\mathbf{AB} \neq \mathbf{BA}$ in general, even if the product exist.
For example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

To distinguish between the product $\mathbf{AB}$ and $\mathbf{BA}$, we say: for $\mathbf{AB}$, $\mathbf{A}$ is pre-multiplied to $\mathbf{B}$; and for $\mathbf{BA}$, $\mathbf{A}$ is post-multiplied to $\mathbf{B}$.

2. $\mathbf{AB} = \mathbf{0}$ does not imply $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$.
i.e. two non-zero matrices can give a zero matrix after multiplication. See the example above.

1.4.7 Properties that are similar to ordinary numbers multiplication

- | | |
|--|---------------------------------|
| 3. $\mathbf{A(BC)} = (\mathbf{AB})\mathbf{C}$ | Associative property |
| 4. $\mathbf{A(B_1 + B_2)} = \mathbf{AB_1 + AB_2}$ | (Left) Distributive property |
| 5. $(\mathbf{C_1 + C_2})\mathbf{A} = \mathbf{C_1A + C_2A}$ | (Right) Distributive property |
| 6. $\mathbf{c(AB)} = (\mathbf{cA})\mathbf{B} = \mathbf{A(cB)}$ | c is any scalar (real number) |

1.4.8 Multiplication with Zero and Identity Matrices

Let $\mathbf{A}$ be an $m \times n$ matrix.

3. $\mathbf{A0_{n \times q} = 0_{m \times q}}$ and $\mathbf{0_{p \times m}A = 0_{p \times n}}$

Zero matrix behaves like number "0" in matrix multiplication

4. $\mathbf{AI_n = I_m A = A}$

Identity matrix behaves like number "1" in matrix multiplication

1.4.9 Powers of Matrices

Let $\mathbf{A}$ be a square matrix and n be a nonnegative integer. We define $\mathbf{A^n}$ as follows:

$$\mathbf{A^n} = \mathbf{AA \dots A} \text{ (n times)}$$

$$\mathbf{A^0} = \mathbf{I}$$

1.4.10 Matrix Transpose

Matrix Transpose of a matrix $\mathbf{A}$, denoted by $\mathbf{A^T}$ (or $\mathbf{A^t}$) interchange the rows and columns of $\mathbf{A}$.

For example,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \quad \mathbf{A^T} = \begin{pmatrix} 1 & 5 \\ 2 & 6 \\ 3 & 7 \\ 4 & 8 \end{pmatrix}$$

If $\mathbf{A}$ is symmetric, then $\mathbf{A} = \mathbf{A^T}$.
--

In other words, the transpose operator does not change a symmetric matrix. For example,

$$\begin{pmatrix} 0 & 1 & 5 \\ 1 & -2 & 3 \\ 5 & 3 & 8 \end{pmatrix}^T = \begin{pmatrix} 0 & 1 & 5 \\ 1 & -2 & 3 \\ 5 & 3 & 8 \end{pmatrix}$$

1.4.11 Some Properties of Transpose

Let $\mathbf{A}$ be an $m \times n$ matrix.

1. $(\mathbf{A}^T)^T = \mathbf{A}$
2. If $\mathbf{B}$ is an $m \times n$ matrix, then $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$.
3. If a is a scalar, then $(a\mathbf{A})^T = a\mathbf{A}^T$.
4. If $\mathbf{B}$ is an $n \times p$ matrix, then $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

Note that, in property 4, the order of the multiplication is “reversed”: On the left side, we pre-multiply $\mathbf{A}$ to $\mathbf{B}$; while on the right hand side, we post multiply $\mathbf{A}^T$ to $\mathbf{B}^T$.

Example:

$$(\mathbf{AB})^T = \left[\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ -1 & -2 & 0 \end{pmatrix} \right]^T = \begin{pmatrix} 2 & 1 & 1 \\ 8 & 7 & 4 \end{pmatrix}^T = \begin{pmatrix} 2 & 8 \\ 1 & 7 \\ 1 & 4 \end{pmatrix}$$
$$\mathbf{B}^T\mathbf{A}^T = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ -1 & -2 & 0 \end{pmatrix}^T \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & -2 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 2 & 8 \\ 1 & 7 \\ 1 & 4 \end{pmatrix}$$

1.5 Matrix Inverses I

The concept of matrix inverse applies strictly to square matrices, and is closely linked to many different topics in linear algebra.

1.5.1 Matrix Equation Form of Linear System

Look at the following linear system:

$$\begin{cases} 4x + 5y + 6z = 1 \\ x - y = 2 \\ y - z = 3 \end{cases} \Leftrightarrow \underbrace{\begin{pmatrix} 4 & 5 & 6 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}}_{\text{Matrix multiplication}} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

We can rewrite the system in matrix equation form as given on the right.

In this form, we have a 3 x 3 matrix pre-multiply to a 3 x 1 column matrix. After performing the matrix multiplication and equating with the right hand side, we recover the original linear system.

More generally, if we have a linear system with m equations and n variables, the matrix equation form will be given as follow.

$$\text{Standard form: } \begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

Matrix Equation Form:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

There are three matrices involved:

On the left we have the m x n matrices made up of all the coefficients of the linear system. We call it the coefficient matrix.

In the middle we have the n x 1 column matrices consisting of the n variables of the system. We call this the variable matrix.

On the right side of the matrix equation is an m x 1 column matrix, consisting of all the constant terms. We call this the constant matrix.

1.5.2 A Concise Notation for Linear System

Matrix equation form allows us to represent linear system symbolically with letters. For example, if we let

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix},$$

then we can represent the linear system as $\mathbf{Ax} = \mathbf{b}$.

We also have a simple way to represent solution of a linear system. Recall this is how we write a solution for the system:

$$x_1 = u_1, x_2 = u_2, \dots, x_n = u_n.$$

In matrix equation form, the solution is represented by an $n \times 1$ column matrix $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$.

Substituting each individual number u_1 to u_n for each variable, can now be replaced collectively by substituting the column matrix $\mathbf{u}$ for the variable matrix $\mathbf{x}$.

Algebraically, we say:

$\mathbf{u}$ is a solution of the linear system $\mathbf{Ax} = \mathbf{b}$, if we have the equality $\mathbf{Au} = \mathbf{b}$.

1.5.3 Solving Matrix Equations

Let $\mathbf{Ax} = \mathbf{b}$ be (the matrix equation of) a linear system. Can we solve the linear system in terms of $\mathbf{A}$ and $\mathbf{b}$?

Let's recall how we solve an ordinary linear equation $ax = b$ with a and b being numbers instead of matrices. If $a \neq 0$, then to solve the equation $ax = b$, we have $x = b/a$.

For our matrix equation $\mathbf{Ax} = \mathbf{b}$, we cannot write: $\mathbf{x} = \mathbf{b}/\mathbf{A}$, as we do not have "division operation" for matrices.

In the number version, we can rewrite the solution $x = b/a = a^{-1}b$ where a^{-1} is called the inverse or reciprocal of the number a .

Can we find an inverse $\mathbf{A}^{-1}$ for matrix $\mathbf{A}$ (which has the similar property as a^{-1})?

The answer is yes, provided $\mathbf{A}$ is a square matrix, and satisfies certain condition.

1.5.4 Inverse of a Square Matrix

Let $\mathbf{A}$ be an $n \times n$ square matrix. The $n \times n$ matrix $\mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$ if the following conditions hold:

$$\mathbf{AA}^{-1} = \mathbf{I} \quad \text{and} \quad \mathbf{A}^{-1}\mathbf{A} = \mathbf{I},$$

Not every square matrix has an inverse. If an inverse of $\mathbf{A}$ exists, we say $\mathbf{A}$ is non-singular or invertible. Otherwise, we say $\mathbf{A}$ is singular or non-invertible.

1.5.5 Inverses of 2 x 2 Matrices

If $\mathbf{A}$ is a 2×2 matrix, there is a simple way to determine whether it is non-singular and to find its inverse.

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

If $ad - bc \neq 0$, then the matrix is non-singular, and the inverse of the matrix is given by this expression:

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$$

If $ad - bc = 0$, then $\mathbf{A}$ is singular. So the inverse of the matrix does not exist.

1.5.6 Example: 2 x 2 Non-Singular Matrix

$$\mathbf{A} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix}$$

We compute $2 \times 3 - (-5) \times (-1) \neq 0$. Therefore the matrix is non-singular.

Using the formula given above, we get the inverse of $\mathbf{A}$:

$$\mathbf{A}^{-1} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$$

You may verify that this matrix is indeed the inverse of $\mathbf{A}$ by checking $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ and $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

1.5.7 Example: Solving a Matrix Equation

Using the matrix $\mathbf{A}$ above, let us set up a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$:

$$\begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

This is the matrix equation form of a linear system with 2 equations and 2 variables.

Now pre-multiplying the inverse matrix $\mathbf{A}^{-1} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ on both sides of the matrix equation:

$$\begin{aligned} \Rightarrow \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 12 \\ 4 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 12 \\ 4 \end{pmatrix} \end{aligned}$$

Hence the solution of the linear system: $x = 12$, $y = 4$.

We have seen this solution is obtained from $\mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 12 \\ 4 \end{pmatrix}$.

1.6 Matrix inverses II

In this section, we discuss matrix inverse for general $n \times n$ square matrices. We will also see how the singularity of a matrix links to other concepts we have introduced so far.

1.6.1 Example: Finding Inverse of 3 x 3 Matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

First we form the 3 x 6 “augmented” matrix using the 3 x 3 identity matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right)$$

We shall now perform Gauss-Jordan elimination on this matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - R_1}} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right)$$

Within the same column, the third type of elementary row operations on different rows can be performed simultaneously.

$$\xrightarrow{R_3 + 2R_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right) \xrightarrow{-R_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right)$$

Though the matrix is already in row echelon form, we need to continue with the G.J.E. steps.

$$\xrightarrow{\substack{R_1 - 3R_3 \\ R_2 + 3R_3}} \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right) \xrightarrow{R_1 - 2R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right)$$

The matrix is now in reduced row echelon form.

In the final matrix, the left half is the identity matrix, and the right half gives the inverse of the original matrix $\mathbf{A}$ as shown below.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right)$$

$\mathbf{I}$ $\mathbf{A}^{-1}$

If we now just focus on the right half of all the augmented matrices in the G.J.E. steps, we observe is that the series of elementary row operations that we performed above will transform the identity matrix to the inverse of our given matrix. (See the next segment)

1.6.2 Finding Inverse of $n \times n$ Matrix

If a sequence of elementary row operations reduce $\mathbf{A}$ to $\mathbf{I}$, the same sequence will also reduce $\mathbf{I}$ to $\mathbf{A}^{-1}$.

$$\begin{array}{ccccccc} \mathbf{A} & \xrightarrow{R_1} & \xrightarrow{R_2} & \dots & \xrightarrow{R_k} & & \mathbf{I} \\ \mathbf{I} & \xrightarrow{R_1} & \xrightarrow{R_2} & \dots & \xrightarrow{R_k} & & \mathbf{A}^{-1} \end{array}$$

This gives us a way to use Gauss-Jordan Elimination to find the inverse of any non-singular matrix.

If the square matrix is singular, i.e. does not have an inverse, we will see in the next segment that there is no sequence of row operations that can reduce the matrix to the identity matrix.

1.6.3 Singularity and Row Echelon Form

Let $\mathbf{A}$ be any square matrix. We can check whether $\mathbf{A}$ is non-singular by looking at its row echelon form:

- (i) If the reduced row-echelon form of $\mathbf{A}$ is an identity matrix, then $\mathbf{A}$ is non-singular.
- (ii) If any row-echelon form of $\mathbf{A}$ has a zero row, then $\mathbf{A}$ is singular.

Here's an example.

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow{\text{G.J.E.}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

After performing Gauss Jordan elimination, we obtain the identity matrix. So we know this matrix is non-singular.

Let's look at another example.

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow{\text{G.E.}} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

It is enough to just perform Gaussian elimination (instead of G.J.E.) to get a row echelon form. As long as we get a zero row, we conclude that the matrix is singular. Note that when there is zero row in the matrix, it is not possible to reduce it further to get the identity matrix.

1.6.4 Matrix Operations on Non-Singular Matrices

Let $\mathbf{A}$, $\mathbf{B}$ be two non-singular matrices (same size) and a is a non-zero scalar.

The table below includes four types of matrix operations: scalar multiplication, transpose, inverse and matrix multiplication (of $\mathbf{A}$ and $\mathbf{B}$).

Matrix	non-singular?	Inverse
$a\mathbf{A}$	yes	$(a\mathbf{A})^{-1} = (1/a)\mathbf{A}^{-1}$
$\mathbf{A}^T$	yes	$(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$
$\mathbf{A}^{-1}$	yes	$(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
$\mathbf{AB}$	yes	$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

This table answers two questions:

The first question (second column) is, if the original matrices $\mathbf{A}$ and $\mathbf{B}$ are non-singular, will the resulting matrices be still non-singular after the matrix operations?

The answer is "yes" for each of the operations. All the resulting matrices have their inverses.

The second question (third column) is, whether we can express the inverses of the resulting matrices in terms of the original ones?

The answer is again "yes". The respective expression for each inverse is given in red in the third column. Note that for matrix multiplication, the inverse of the product is equal to the product of the inverses with the order reversed.

1.6.5 Example: Matrix Operations

$$\mathbf{A} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{with inverses} \quad \mathbf{A}^{-1} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} \quad \mathbf{B}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(i) $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$

$$\mathbf{A}^T = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$$

$$(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$$

To find the inverse of $\mathbf{A}^T$, we can either do it directly using the formula, or just use the fact that this is the same as the transpose of $\mathbf{A}^{-1}$.

(ii) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

$$\mathbf{AB} = \begin{pmatrix} 2 & -5 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} -1 & -2 \\ 3 & 5 \end{pmatrix}$$

To find the inverse of $\mathbf{AB}$, instead of working out its product and take the inverse, we can use the product of the inverses of $\mathbf{B}$ and $\mathbf{A}$.

1.6.6 Multiplications of Non-Singular Matrices

Given $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k$ are all non-singular matrices of the same size.

The product $\mathbf{A}_1\mathbf{A}_2\dots\mathbf{A}_k$ is a non-singular matrix and the inverse of $\mathbf{A}_1\mathbf{A}_2\dots\mathbf{A}_k$ is

$$(\mathbf{A}_1\mathbf{A}_2\dots\mathbf{A}_k)^{-1} = (\mathbf{A}_k)^{-1} \dots (\mathbf{A}_2)^{-1}(\mathbf{A}_1)^{-1}$$

Note that the order of the matrices is reversed.

If $\mathbf{A}$ is non-singular, then we can define its negative power:

$$(\mathbf{A})^{-n} = (\mathbf{A}^{-1})^n = \underbrace{\mathbf{A}^{-1}\mathbf{A}^{-1} \dots \mathbf{A}^{-1}}_{n \text{ times}}$$

1.6.7 Singularity and Solutions of Linear System

If $\mathbf{A}$ is non-singular,

- i. The linear system $\mathbf{Ax} = \mathbf{b}$ has exactly one solution given by $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.
- ii. The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.

If $\mathbf{A}$ is singular,

- i. The linear system $\mathbf{Ax} = \mathbf{b}$ has either no solution or infinitely many solutions.
- ii. The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has both the trivial solution and non-trivial solutions.

1.6.8 Example: Solutions of Linear System

We use two earlier examples:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow{G.J.E.} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow{G.E.} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Recall that $\mathbf{A}$ is non-singular and $\mathbf{B}$ is singular.

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{Using } \mathbf{A} \text{ to form the homogeneous system } \mathbf{Ax} = \mathbf{0}, \text{ we can tell it has only the trivial solution.}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix} \quad \text{If } \mathbf{A} \text{ is the coefficient matrix of a non-homogeneous system, we can tell that it has exactly one solution.}$$

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{Using } \mathbf{B} \text{ to form the homogeneous system } \mathbf{Bx} = \mathbf{0}, \text{ we can tell it has infinitely many solutions.}$$

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix} \quad \text{If } \mathbf{B} \text{ is the coefficient matrix of a non-homogeneous system, we cannot tell the number of solutions: it can have either no solution or infinitely many solutions (i.e. it cannot have only one solution).}$$

In this case, we need to apply Gaussian elimination to the augmented matrix and look at the row echelon form.

1.7 Determinants of Matrices

Determinant is another useful concept associated with a square matrix. We shall see in this section how to compute determinant, some of its basic properties, and how it is related to the singularity of a matrix.

1.7.1 Determinant of 2 x 2 Matrices

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a 2 x 2 matrix.

Then the determinant of $\mathbf{A}$, denoted by $\det(\mathbf{A})$, is the number defined as $\det(\mathbf{A}) = ad - bc$.

An alternative notation for $\det(\mathbf{A})$ is: $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$

So if $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, then $\det(\mathbf{A}) = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \times 4 - 2 \times 3 = -2$.

From earlier section, we have seen that this number is related to the singularity of the matrix:

If $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is **non-singular**.

If $\det(\mathbf{A}) = 0$, then $\mathbf{A}$ is **singular**.

In general, how do we find the determinant of an $n \times n$ square matrix? There are in fact various ways to compute determinants. A systematic but tedious way to do it is by induction (expansion).

1.7.2 Determinant of 3 x 3 Matrices

$$\mathbf{B} = \begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix}$$

We shall compute the 3 x 3 determinant of $\mathbf{B}$ in terms of 2 x 2 determinants:

$$\det(\mathbf{B}) = (-3) \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} - (-2) \begin{vmatrix} 4 & 1 \\ 0 & 4 \end{vmatrix} + 4 \begin{vmatrix} 4 & 3 \\ 0 & 2 \end{vmatrix}$$

The 3 coefficients (-3, -2, 4) attached to the 2 x 2 determinants come from the first row of $\mathbf{B}$.

The 2x2 determinants are all parts of the matrix $\mathbf{B}$ (we call them the submatrices of $\mathbf{B}$):

The first one $\begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix}$ is obtained from $\mathbf{B}$ by removing its first row and first column.

The second one $\begin{vmatrix} 4 & 1 \\ 0 & 4 \end{vmatrix}$ is obtained from $\mathbf{B}$ by removing its first row and second column.

The third one $\begin{vmatrix} 4 & 3 \\ 0 & 2 \end{vmatrix}$ is obtained from $\mathbf{B}$ by removing its first row and third column.

Note that, on the right hand side, we are taking the alternating sum. In other words, we have the first term minus the second term plus the third term. i.e. the terms alternate between + and - sign, starting with a + sign.

We expand the 2 x 2 determinants further to obtain the final answer:

$$\det(\mathbf{B}) = -3(3 \times 4 - 1 \times 2) + 2(4 \times 4 - 1 \times 0) + 4(4 \times 2 - 3 \times 0) = 34.$$

1.7.3 Determinant of $n \times n$ Matrices

In general, an $n \times n$ determinant is computed in terms of $(n-1) \times (n-1)$ determinants. We shall illustrate using a 4×4 matrix:

$$\mathbf{C} = \begin{pmatrix} 0 & -1 & 2 & 0 \\ 2 & -3 & 3 & -2 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & -1 \end{pmatrix}$$

The determinant is given as the expansion of four 3×3 determinants:

$$\det(\mathbf{C}) = 0 \begin{vmatrix} -3 & 3 & -2 \\ 2 & 4 & 0 \\ 0 & 2 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 2 & 3 & -2 \\ 0 & 4 & 0 \\ 0 & 2 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -3 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{vmatrix} - 0 \begin{vmatrix} 2 & -3 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 2 \end{vmatrix}$$

Like before, the coefficients indicated in red come from the first row of $\mathbf{C}$. The 3×3 submatrices are obtained by deleting the first row, and one of the 4 columns of $\mathbf{C}$. Finally, we apply the alternating sum to the 4 terms.

The four 3×3 determinants on the right hand side can be further expanded in terms of 2×2 determinants in a similar way. I will leave it to the readers to verify that the final answer is

$$\det(\mathbf{C}) = -16.$$

This inductive method is quite tedious when the size of the matrix is large. We shall see later how to make use of some properties of determinant to simplify the computation.

1.7.4 Determinant and Singularity

The relation between determinant and singularity holds for any $n \times n$ square matrix $\mathbf{A}$:

- if $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is non-singular;
- if $\det(\mathbf{A}) = 0$, then $\mathbf{A}$ is singular.

$$\mathbf{B} = \begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix}$$

We have seen before that $\det(\mathbf{B}) = 34 \neq 0$, so $\mathbf{B}$ is non-singular.

$$\mathbf{C} = \begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 1 & 1 & 5 \end{pmatrix}$$

$\det(\mathbf{C}) = 0$ (check!), so $\mathbf{C}$ is singular.

1.7.5 Determinants of Triangular Matrices

If $\mathbf{A}$ is a triangular matrix or diagonal matrix, then the determinant of $\mathbf{A}$ is equal to the product of the diagonal entries of $\mathbf{A}$.

Look at the three matrices below. They are upper triangular, lower triangular and diagonal matrices respectively.

$$\begin{vmatrix} -1 & 3.5 & 14 \\ 0 & 5 & -26 \\ 0 & 0 & 2 \end{vmatrix} \quad \begin{vmatrix} -2 & 0 & 0 \\ 99 & 0 & 0 \\ 10 & -4.5 & 10 \end{vmatrix} \quad \begin{vmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= (-1) \times 5 \times 2 = -10 \quad = (-2) \times 0 \times 10 = 0 \quad = 3 \times 4 \times 2 = 24$$

So we know that the first and third matrices are non-singular as their determinants are not 0, while the second matrix is singular.

1.7.6 Determinants with Two Identical Rows (or Columns)

1. The determinant of a square matrix with two identical rows is zero.
2. The determinant of a square matrix with two identical columns is zero.

$$\begin{pmatrix} 4 & -2 \\ 4 & -2 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 4 \\ -1 & 10 & 4 \\ 1 & 2 & 4 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & -3 & -3 & 9 \\ 2 & 4 & 4 & 0 \\ 0 & -2 & -2 & -1 \end{pmatrix}$$

All the three matrices above have either two identical rows or two identical columns. So their determinants are known to be 0 (without having to expand in terms of smaller determinants) and they are all singular matrices.

1.7.7 Determinants and Elementary Row Operations

The table below summarises the effect of the three types of elementary row operations on determinants.

E.R.O	Determinant
$\mathbf{A} \xrightarrow{kR_i} \mathbf{B}$	$\det(\mathbf{B}) = k \det(\mathbf{A})$
$\mathbf{A} \xrightarrow{R_i \leftrightarrow R_j} \mathbf{B}$	$\det(\mathbf{B}) = -\det(\mathbf{A})$
$\mathbf{A} \xrightarrow{R_i + kR_j} \mathbf{B}$	$\det(\mathbf{B}) = \det(\mathbf{A})$

An example for the first type of row operation:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix} \quad \text{Observe that } \mathbf{B} \text{ is obtained from } \mathbf{A} \text{ by multiplying the second row by 3, i.e. } 3R_2.$$

$\det(\mathbf{B}) = 3 \det(\mathbf{A})$

An example for the second type of row operation:

$$\mathbf{A} = \begin{pmatrix} 3 & 3 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} \quad \text{Observe that } \mathbf{B} \text{ is obtained from } \mathbf{A} \text{ by interchanging first and third row of } \mathbf{A}, \text{ i.e. } R_1 \leftrightarrow R_3.$$

$\det(\mathbf{B}) = -\det(\mathbf{A})$

The third type of row operation does not change the determinant. We shall see an example in the next segment.

1.7.8 Find Determinants by Gaussian Elimination

Instead of computing the 4 x 4 determinant below inductively, we are going to perform elementary row operations to reduce the matrix to an upper triangular matrix.

$$\begin{vmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_2 - R_1} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_4 - 2R_3} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix}$$

The first and third row operations above are of the third type, and hence do not affect the determinant. The second row operations are of the second type, and the determinants will differ by a factor of -1.

We observe that the last determinant on the right is in row echelon form (an upper triangular matrix). So the determinant can be obtained easily from the diagonal entries 3, 2, 1 and -1. Hence the original determinant is given by $(3 \times 2 \times 1 \times -1) = -6$.

1.7.9 Determinant and Matrix Operations

The last thing we want to discuss is the effect of some matrix operations on the determinants.

Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ square matrices, and c be a scalar.

$$1. \det(c\mathbf{A}) = c^n \det(\mathbf{A})$$

(The power n in statement 1 is the size of $\mathbf{A}$)

$$2. \det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$$

$$3. \det(\mathbf{A}^T) = \det(\mathbf{A})$$

$$4. \text{ if } \mathbf{A} \text{ is non-singular, then } \det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$$

(The two sides in statement 5 are not equal in general. For some special cases, such as one of the matrices is zero, equality can still hold)

$$5. \det(\mathbf{A} + \mathbf{B}) \neq \det(\mathbf{A}) + \det(\mathbf{B})$$

1.7.10 Example: Matrix Operations

Let's demonstrate the above properties using our two previous matrices $\mathbf{B}$ and $\mathbf{C}$:

$$\mathbf{B} = \begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 1 & 1 & 5 \end{pmatrix}$$

$$\det(\mathbf{B}) = 34 \quad \det(\mathbf{C}) = 0$$

Example 1 $\det(2\mathbf{B}) = 2^3 \det(\mathbf{B})$

$$\begin{vmatrix} -6 & -4 & 8 \\ 8 & 6 & 2 \\ 0 & 4 & 8 \end{vmatrix} = 2^3 \times 34 = 272$$

Example 2 $\det(\mathbf{B}^T) = \det(\mathbf{B})$

$$\begin{vmatrix} -3 & 4 & 0 \\ -2 & 3 & 2 \\ 4 & 1 & 4 \end{vmatrix} = 34$$

Example 3 $\det(\mathbf{BC}) = \det(\mathbf{B})\det(\mathbf{C})$

$$\det\left[\begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix} \begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 1 & 1 & 5 \end{pmatrix}\right] = 34 \times 0 = 0$$

1.8 Ranks of Matrices

In this short section, we introduce another commonly used concept: the rank of a matrix. We will establish the link between rank of a matrix and some other concepts that we have discussed so far.

1.8.1 Rank of a Matrix

Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{R}$ be a row echelon form of $\mathbf{A}$.

$$\mathbf{A} \xrightarrow{\text{G. E.}} \mathbf{R} \text{ (REF)}$$

The rank of $\mathbf{A}$ is the number of nonzero rows of $\mathbf{R}$.

Notation: $\text{rank}(\mathbf{A})$

It follows that

$\text{rank}(\mathbf{A}) = \text{the number of leading entries in } \mathbf{R} = \text{the number of pivot columns in } \mathbf{R}$

1.8.2 Example: Rank of a Matrix

The rank of the following 6×4 matrix $\mathbf{A}$ is 3, i.e. $\text{rank}(\mathbf{A}) = 3$, by counting the number of nonzero rows or number of leading entries or number of pivot columns of the row echelon form on the right.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Pivot Columns

Non-zero rows

Zero matrix: $\mathbf{0} = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$

$\text{rank}(\mathbf{0}) = 0$, as there is no non-zero row.

In fact, zero matrices are the only matrices that have rank equal to 0.

Identity matrix: $\mathbf{I}_n = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{pmatrix}$

$\text{rank}(\mathbf{I}_n) = n$, as every row is a non-zero row.

Note that both matrices are already in row echelon form. So there is no need to perform Gaussian Elimination.

1.8.3 Largest Possible Rank of a Matrix

What is the largest possible rank of a 5×3 matrix $\mathbf{A}$?

Since $\mathbf{A}$ has 3 columns, and the rank is given by the number of pivot columns in the row echelon form of $\mathbf{A}$, so the largest possible rank of $\mathbf{A}$ is 3.

What is the largest possible rank of a 3×5 matrix $\mathbf{B}$?

Since $\mathbf{B}$ has 3 rows, and the rank is given by the number of nonzero rows in the row echelon form of $\mathbf{B}$, so the largest possible rank of $\mathbf{B}$ is 3.

1.8.4 Full Rank

From the above discussion, we notice that the possible rank of a matrix is bounded by the smaller of the number of rows and the number of columns of the matrix. So for an $m \times n$ matrix $\mathbf{A}$,

$$\text{rank}(\mathbf{A}) \leq \min\{m, n\}.$$

the smaller of the
two numbers m and n

For example, if $\mathbf{A}$ is a 4×6 general matrix, the largest possible rank it can have is 4, and the possible rank of $\mathbf{A}$ is 0, 1, 2, 3 and 4. Note that the rank can only be 0 if $\mathbf{A}$ is the 4×6 zero matrix.

When a matrix $\mathbf{A}$ achieves its largest possible rank, i.e. $\text{rank}(\mathbf{A}) = \min(m, n)$, we say that $\mathbf{A}$ is a full rank matrix.

For example, if a 4×6 matrix $\mathbf{A}$ has a rank of 4, then $\mathbf{A}$ is a full rank matrix.

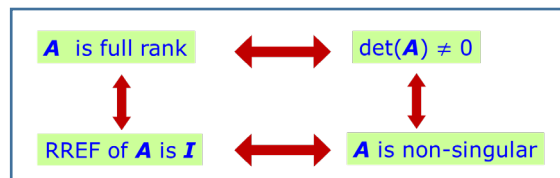
The $n \times n$ identity matrix $\mathbf{I}_n$ has rank equal to n . So $\mathbf{I}_n$ is a full rank matrix.

On the other hand, the 3×3 matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ has rank 1. So it is not full rank.

1.8.5 Rank and Determinant

The following discussion links the concept of rank with some of the results we have mentioned in earlier sections.

An $n \times n$ square matrix $\mathbf{A}$ is full rank means $\text{rank}(\mathbf{A}) = n$. This means the reduced row echelon form of $\mathbf{A}$ has n pivot columns. So the reduced row echelon form of $\mathbf{A}$ must be the identity matrix $\mathbf{I}_n$. This in turn means $\mathbf{A}$ is non-singular and $\det(\mathbf{A}) \neq 0$.



We say, any one of these conditions on the matrix $\mathbf{A}$ is equivalent to the other three.

Here is a concrete 3×3 matrix: $\begin{pmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{pmatrix}$

The following diagram connects the various conditions on the matrix:

